

Transcriptome landscape of a bacterial pathogen under plant immunity

Tatsuya Nobori^a, André C. Velásquez^b, Jingni Wu^{a,1}, Brian H. Kvitko^b, James M. Kremer^{b,2}, Yiming Wang^a, Sheng Yang He^{b,c,3}, and Kenichi Tsuda^{a,3}

^aDepartment of Plant Microbe Interactions, Max Planck Institute for Plant Breeding Research, 50829 Cologne, Germany; ^bMichigan State University-Department of Energy Plant Research Laboratory, Michigan State University, East Lansing, MI 48824; and ^cHoward Hughes Medical Institute, Michigan State University, East Lansing, MI 48824

Contributed by Sheng Yang He, February 2, 2018 (sent for review January 12, 2018; reviewed by Jeffery L. Dangl and Steven E. Lindow)

Plant pathogens can cause serious diseases that impact global agriculture. The plant innate immunity, when fully activated, can halt pathogen growth in plants. Despite extensive studies into the molecular and genetic bases of plant immunity against pathogens, the influence of plant immunity in global pathogen metabolism to restrict pathogen growth is poorly understood. Here, we developed RNA sequencing pipelines for analyzing bacterial transcriptomes *in planta* and determined high-resolution transcriptome patterns of the foliar bacterial pathogen *Pseudomonas syringae* in *Arabidopsis thaliana* with a total of 27 combinations of plant immunity mutants and bacterial strains. Bacterial transcriptomes were analyzed at 6 h post infection to capture early effects of plant immunity on bacterial processes and to avoid secondary effects caused by different bacterial population densities *in planta*. We identified specific “immune-responsive” bacterial genes and processes, including those that are activated in susceptible plants and suppressed by plant immune activation. Expression patterns of immune-responsive bacterial genes at the early time point were tightly linked to later bacterial growth levels in different host genotypes. Moreover, we found that a bacterial iron acquisition pathway is commonly suppressed by multiple plant immune-signaling pathways. Overexpression of a *P. syringae* sigma factor gene involved in iron regulation and other processes partially countered bacterial growth restriction during the plant immune response triggered by AvrRpt2. Collectively, this study defines the effects of plant immunity on the transcriptome of a bacterial pathogen and sheds light on the enigmatic mechanisms of bacterial growth inhibition during the plant immune response.

plant immunity | type III effector | microbiome | plant hormone | iron response

Photosynthate-rich plants provide an important growth niche for microbes including bacterial pathogens. In most cases, the plant immune system can effectively monitor microbial colonization and restrict the growth of a great majority of them (1). Excessive proliferation of evolved pathogenic microbes, however, can lead to plant diseases. Understanding how plant diseases occur and how the plant immune system effectively monitors and restricts pathogen infection represents one of the most important research areas in plant science.

Molecular genetic studies have revealed a repertoire of plant immune receptors and downstream immune-signaling components and their mode of actions within plant cells. As the first layer of the plant immune surveillance system, cell-surface pattern-recognition receptors recognize conserved molecular features of microbes such as bacterial flagellin and fungal cell wall components (chitin) to activate pattern-triggered immunity (PTI) (2, 3). To cause disease in plants, bacterial pathogens have evolved a battery of virulence molecules including type III effectors (T3Es) to subvert PTI (4). Plants, in turn, have evolved intracellular nucleotide-binding domain and leucine-rich repeat (NLR)-containing receptors, which recognize individual T3Es, as a second layer of the plant immune surveillance system, leading to effector-triggered immunity (ETI) (5). Although PTI and ETI receptors show distinctive cellular

locations and biochemical properties, they share some downstream signaling components, such as signaling pathways mediated by the defense hormones salicylic acid (SA), jasmonate (JA), and ethylene (ET) (6). Despite extensive studies on the molecular, biochemical, and transcriptomic reprogramming within plant cells during immune activation, little is known about how plant immunity influences bacterial cellular processes and ultimately suppresses bacterial growth.

Previous studies have revealed bacterial processes that are important for their growth *in planta*. These include siderophores, exopolysaccharides, quorum sensing, multidrug resistance efflux pumps, plant hormones and their mimics, and toxins as well as T3Es secreted into the plant cell via the type III secretion system (T3SS) (4, 7–12). A number of T3Es suppress plant immune signaling as an important mechanism of bacterial virulence (13). T3Es have also been shown to manipulate plant intercellular (apoplastic) conditions to favor their proliferation. For instance, *Xanthomonas oryzae* uses T3Es, which function as transcriptional activators, to directly induce sugar-transporter genes, thereby

Significance

Plants have evolved a powerful innate immune system to defend against microbial pathogens. Despite extensive studies, how plant immunity ultimately inhibits bacterial pathogen growth is largely unknown, due to difficulties in profiling bacterial responses *in planta*. In this study, we established two methods for *in planta* bacterial transcriptome analysis using RNA sequencing. By analyzing 27 combinations of plant immunity mutants and *Pseudomonas syringae* strains, we succeeded in the identification of specific bacterial transcriptomic signatures that are influenced by plant immune activation. In addition, we found that overexpression of an immune-responsive *P. syringae* sigma factor gene involved in iron regulation could partially counter bacterial growth restriction during plant immunity. This study illuminates the enigmatic mechanisms of bacterial growth inhibition by plant immunity.

Author contributions: T.N., S.Y.H., and K.T. designed research; T.N., A.C.V., J.W., B.H.K., J.M.K., and Y.W. performed research; T.N. and K.T. analyzed data; and T.N., S.Y.H., and K.T. wrote the paper.

Reviewers: J.L.D., University of North Carolina; and S.E.L., University of California.

Conflict of interest statement: J.K. and J.L.D. are coauthors on a 2017 Perspective article.

Published under the PNAS license.

Data Deposition: The RNA-sequencing data used in this study are deposited in the National Center for Biotechnology Information Gene Expression Omnibus database (accession no. GSE103442).

¹Present addresses: National Key Laboratory of Plant Molecular Genetics, Shanghai Institute of Plant Physiology and Ecology, Chinese Academy of Sciences, 200032 Shanghai, China.

²Present address: AgBiome, Research Triangle Park, NC 27709.

³To whom correspondence may be addressed. Email: hes@msu.edu or tsuda@mpipz.mpg.de.

This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1800529115/-DCSupplemental.

Published online March 12, 2018.